

Project Design Phase-II
Data Flow Diagram & User Stories

Date	29 JUNE 2026
Team ID	
Project Name	Human Development Index (HDI) Predictor
Maximum Marks	4 Marks

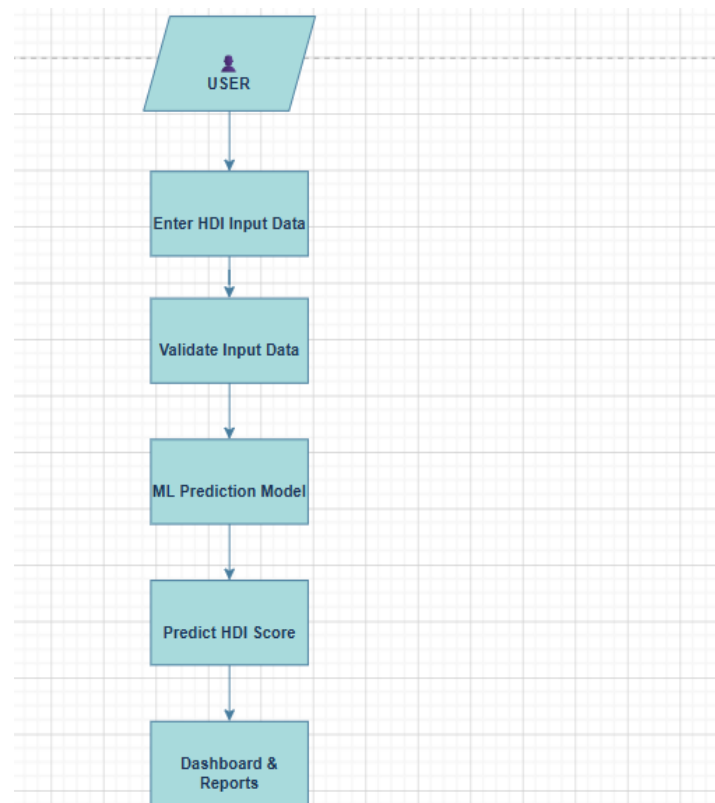
Data Flow Diagrams:

A **Data Flow Diagram (DFD)** is a graphical representation that illustrates how data moves through the HDI Predictor System. It clearly depicts the flow of information from user input to prediction generation, including data processing, storage, and output generation. The DFD helps visualize the overall workflow of the system and demonstrates how each component interacts to produce accurate **Human Development Index (HDI)** predictions.

REFERENCE: [SIMPLE DIAGRAM](#)

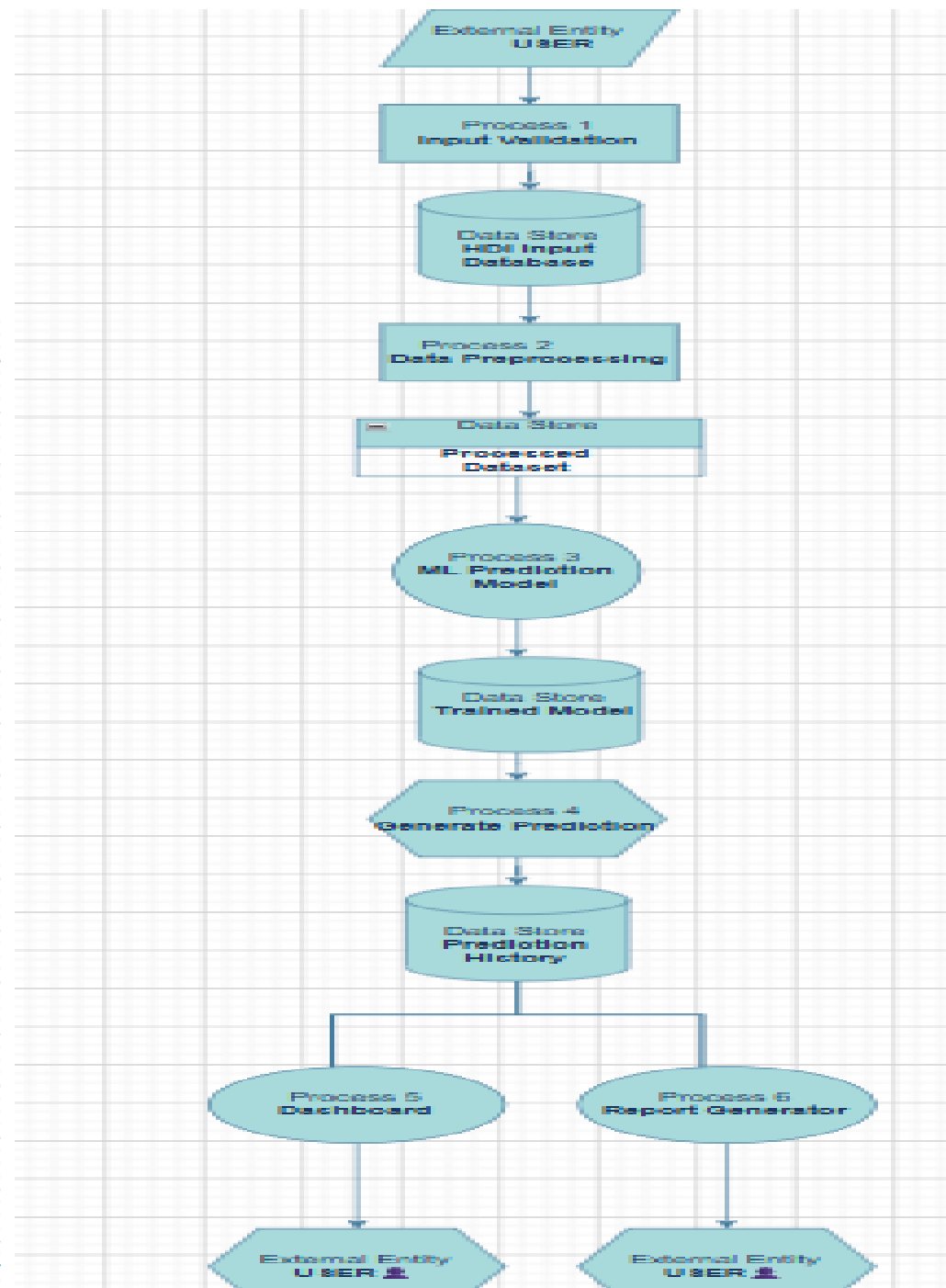
DFD Level-0 (Industry Standard)

REFERENCE LINK FOR DFD LEVEL 0 : [SIMPLEDIAGRAM](#)



Simplified Data Flow Explanation

1. The User enters the required HDI input parameters.
2. The system validates the input data to ensure accuracy and completeness.
3. The validated data is processed by the Machine Learning Prediction Model.
4. The model generates the predicted HDI score.
5. The prediction results are displayed through the Dashboard, and reports can be generated for further analysis.



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Government Policymaker	User Authentication	USN-1	As a policymaker, I can register for the HDI Predictor application using my email and password.	User can successfully create an account.	High	Sprint-1
Government Policymaker	User Authentication	USN-2	As a policymaker, I can log in securely using my registered credentials.	User can log in and access the dashboard.	High	Sprint-1
Government Policymaker	HDI Prediction	USN-3	As a policymaker, I can enter HDI indicators for prediction.	System accepts valid input values.	High	Sprint-1
Government Policymaker	Data Validation	USN-4	As a policymaker, I want the system to validate all input values before prediction.	Invalid or missing values are detected with appropriate messages.	High	Sprint-1
Government Policymaker	Machine Learning Prediction	USN-5	As a policymaker, I can generate an HDI prediction using the trained ML model.	HDI prediction is generated successfully.	High	Sprint-1
Government Policymaker	Dashboard	USN-6	As a policymaker, I can view the predicted HDI score on an interactive dashboard.	Dashboard displays prediction and key metrics	High	Sprint-1
Government Policymaker	Data Visualization	USN-7	As a policymaker, I can visualize HDI trends and comparisons using charts.	Charts are displayed correctly.	Medium	Sprint-2
Government Policymaker	Report Generation	USN-8	As a policymaker, I can download prediction reports in PDF format.	Report is generated and downloaded successfully.	Medium	Sprint-2
Government Policymaker	Prediction History	USN-9	As a policymaker, I can view previous prediction records.	Prediction history is available and searchable.	Medium	Sprint-2

Administrator	Dataset Management	USN-10	As an administrator, I can upload and manage HDI datasets used for training.	Dataset upload and management are successful.	High	Sprint-2
Administrator	Model Management	USN-11	As an administrator, I can retrain or update the machine learning model.	Model training completes successfully and is saved.	Medium	Sprint-3
Administrator	User Management	USN-12	As an administrator, I can manage user accounts and system access.	User accounts can be viewed, updated, or removed.	Medium	Sprint-3